Milestone 5 Progress Evaluation

Project Ragnarok: Post-Quantum Cybersecurity with Lightning Data

Group Members

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Advisor / Client

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Institution: Florida Institute of Technology – Senior Design

1. **Progress of Current Milestone (Progress Matrix)**

Task	Completion	Aidan	Gianni	Joanna	To Do
1. API Authentication and Integration	70%	15%	45%	10%	Complete OAuth2/JWT, rate limiting, and Pi endpoint verification.
2. Entropy Assessment and Research Website	80%	60%	10%	10%	Integrate and test a NIST-approved PQC scheme; entropy extraction is supporting work.
3. Database Security and Raspberry Pi Deployment	70%	30%	20%	20%	Finish deployment verification of restricted DB roles, TLS, telemetry encryption, and scrubbed audit logs.

2. **Discussion of Accomplished Tasks**

Task 1: API Authentication and Integration

Implemented bearer-token authentication, role-based endpoint authorization, authentication database migrations, browser login/dashboard flows, and an authentication verification script.

Joanna developed the authentication and access-control foundation; Aidan integrated the branches and browser workflows; Gianni developed the Python API structure and corrected Rust/Docker build issues. OAuth2/JWT and rate limiting remain to be completed. The Raspberry Pi has the latest main branch loaded, as confirmed by the team; installation alone does not establish that every endpoint or security feature has passed hardware testing.

Task 2: Entropy Assessment and Research Website

Aidan implemented entropy-source assessment and ingestion tooling, preserved the lightning assessment work, and integrated the research website with the main application. SHA3-based research output and recoverable file-encryption workflows were integrated into main. Additional development includes researcher accounts, private workspaces, a live environmental dashboard, and OS-seeded Toeplitz extraction with public seed and source provenance; these newer features are on the open research branch (PR #8) and are not included in the confirmed main-branch Pi deployment. The earlier preliminary entropy estimate does not by itself establish a validated joint source bound or a guaranteed 256-bit secret key. NIST-approved PQC integration remains unfinished; SHA3, AES, and Toeplitz extraction do not substitute for that milestone.

Task 3: Database Security and Raspberry Pi Deployment

Implemented parameterized database queries, authentication-related schema changes, input validation, and application-level access controls. Joanna led database access-control and security preparation, including the restricted application-role work reported by the team. Aidan contributed application hardening and secure integration; Gianni supported deployment on the Raspberry Pi and got the automatic data importers working. Database TLS enforcement, field-level telemetry encryption, restricted-role deployment, and log scrubbing require final implementation or deployment verification before end-to-end security can be marked complete.

3. Team Member Contributions

Aidan Nelappana: Implemented entropy assessment and ingestion improvements, application hardening, and integration of the authentication, Python API, and website branches. Added browser authentication integration, recoverable file encryption, and research visualizations to the integrated application. Further development includes researcher accounts and private workspaces, user guidance and a demo deck, automated environmental collection and a live dashboard, and an OS-seeded Toeplitz research extractor with downloadable provenance. These newer research-branch features remain separate from the main branch currently loaded on the Pi. Led the entropy research implementation and documented remaining source-validation assumptions.

Gianni Bubb: Developed and maintained the Python API branch, separated Pydantic models, updated API documentation, and resolved Rust/Docker build issues. Worked on Raspberry Pi deployment and got the automatic data importers working. The team confirms that the Raspberry Pi now has the latest main branch loaded. OAuth2/JWT work has primarily involved research, with implementation still outstanding; full website/API verification on the Pi remains a next-milestone task.

Joanna Zhang: Developed the authentication and access-control foundation, including database migrations, Rust authentication and role checks, protected endpoints, browser login/dashboard scripts, input validation, and an authentication verification script. Led database security preparation and contributed parameterized-query and restricted application-role work. Continued work covers database TLS enforcement and field-level telemetry encryption, with final deployment verification still required. Earlier authentication work was integrated into main for this milestone.

4. Plan for Next Milestone (Task Matrix)

Task	Aidan	Gianni	Joanna
1. Complete Authentication and PQC Integration	Key pipeline/PQC	Python API/JWT	RBAC/auth tests
2. Finalize Entropy Research and Validation	Lead assessment	Pi data collection	Dataset validation
3. Validate Pi Deployment and End-to-End Security	Integration fixes	Lead Pi testing	Lead DB security
4. Finish GUI, Documentation, and Demonstration	GUI/manual/demo	Setup/runbook	Usability/review

5. Discussion of Planned Tasks

Task 1: Complete Authentication and PQC Integration

Complete OAuth2/JWT authentication and rate limiting in the Python API, integrate a NIST-approved PQC scheme with the key-generation workflow, and preserve the browser authentication flows. Aidan leads the key pipeline and PQC integration, Gianni leads Python API/JWT implementation, and Joanna verifies roles and authorization. Complete when an authenticated client can run the documented key workflow and automated checks cover invalid, expired, and unauthorized credentials, role restrictions, and PQC interoperability.

Task 2: Finalize Entropy Research and Validation

Finalize the lightning assessment and SHA3/Toeplitz research workflow, including provenance, input consumption, reproducibility, and explicit entropy assumptions. Aidan leads assessment and extractor validation, Gianni supports fresh data acquisition on the Pi, and Joanna checks dataset structure and access. Complete when the assessment can be reproduced from documented

inputs, remaining physical-source and dependence limitations are stated, and the research output is clearly distinguished from secret-key security claims.

Task 3: Validate Pi Deployment and End-to-End Security

Verify the latest main branch already loaded on the Raspberry Pi, integrate reviewed remaining features, and resolve website routing, startup, importer, and recovery problems. Gianni leads hardware testing, Joanna leads restricted database-role, TLS, telemetry-encryption, and scrubbed audit-log verification, and Aidan resolves integration defects. Include failed-login monitoring, API load/latency measurements, and a sustained reliability run. Complete when the Pi can restart and serve the documented workflow, access-control checks pass, and benchmark/security results and remaining limitations are recorded.

Task 4: Finish GUI, Documentation, and Demonstration

Finish and integrate the accounts, private workspaces, research dashboard, and extraction GUI after review. Aidan leads GUI integration, the manual, and the demonstration; Gianni documents Pi installation, startup, and recovery; Joanna reviews usability and security guidance. Complete when a non-programmer can follow the guide to log in and complete the supported workflow, the final report and test evidence are consistent with the delivered build, and the team can demonstrate the project end to end.

Meetings

Client Meeting Date(s)

Oct 19, 2026

Client Feedback

Faculty Advisor Meeting Date(s)

Oct 19, 2026

Faculty Advisor Feedback

Task 1:

Task 2:

Task 3:

Task 4:

Faculty Advisor Signature: _____ Date: _____

Evaluation by Faculty Advisor

- Faculty Advisor: detach and return this page to Dr. Chan (HC 209) or email the scores to pkc@cs.fit.edu
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Joanna Zhang	0	1	2	3	4	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10
Gianni Bubb	0	1	2	3	4	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10
Aidan Nelappana	0	1	2	3	4	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10

- Score (0-10) for each member: circle a score (or circle two adjacent scores for .25 or write down a real number between 0 and 10)

Faculty Advisor Signature: _____ Date: _____